Lagrange interpolation and finite differences

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L5-X4 Level 5 Advanced · Algebra · Beyond the textbook

Working with the values of a polynomial instead of its coefficients, namely the interpolation formula and uniqueness, finite differences and the degree, the alternating binomial identity, values at integers and contest uses modulo a prime.
By the end you can use Lagrange interpolation, its uniqueness and its leading-coefficient form to evaluate polynomials and symmetric sums; use finite differences and the alternating binomial identity for polynomials of degree less than n to find P(n+1) and to bound values; decide divisibility of P(n) for all integers n from finitely many values, and to use these tools modulo a prime.
Prerequisites: Polynomials over the integers (L5-AL1). **Contests:** BMO Round 2, EGMO and IMO selection, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

A polynomial P of degree at most 3 satisfies $P(0) = 1$, $P(1) = 3$, $P(2) = 9$, $P(3) = 27$. Find $P(4)$.

Solving four linear equations for the coefficients works, but there is no need to find them. Write the values in a row and take differences repeatedly:

$$\begin{array}{ccccccccc} 1 & & 3 & & 9 & & 27 & & \mathbf{65} \\ & 2 & & 6 & & 18 & & \mathbf{38} & \\ & & 4 & & 12 & & \mathbf{20} & & \\ & & & 8 & & \mathbf{8} & & & \end{array}$$

A cubic has constant third differences, so the bottom row continues with 8, and adding back up the table gives $P(4) = 65$. The values of a polynomial at a few points carry the same information as its coefficients, and many contest problems are easy in the values and hard in the coefficients. This unit is about that point of view, namely Lagrange's interpolation formula (published by Waring in 1779 and by Lagrange in 1795), the finite differences that go back to Newton, the alternating binomial identity that combines them, and their uses with integers. School courses treat polynomials through their coefficients, which is why these tools are missed. We assume L5-AL1, in particular the binomial basis $\binom{x}{k}$ and its Theorem 3, which says $P(x) = \sum_k c_k \binom{x}{k}$ with $c_k = \Delta^k P(0)$, and P takes integer values at all integers if and only if every c_k is an integer.

1.1 Lagrange interpolation

Theorem 1.
Let $x_0, \ldots, x_n$ be distinct and $y_0, \ldots, y_n$ arbitrary. There is exactly one polynomial P of degree at most n with $P(x_i) = y_i$ for all i, namely

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$$P(x) = \sum_{i=0}^{n} y_i L_i(x), \qquad L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

Proof. L_i has degree n, $L_i(x_i) = 1$ and $L_i(x_k) = 0$ for $k \neq i$, so P works. If P_1, P_2 both work, $P_1 - P_2$ has degree at most n and $n+1$ roots, so $P_1 - P_2$ is zero.

Uniqueness is used at least as often as the formula. To show two polynomials of degree at most n are equal, check them at $n+1$ points. Reading off the coefficient of x^n in each L_i gives the form of the formula used most in olympiads.

Theorem 2.
Let $x_1, \ldots, x_n$ be distinct, $Q(x) = \prod_j (x - x_j)$, so that $Q'(x_i) = \prod_{j \neq i}(x_i - x_j)$. For every polynomial R of degree at most $n-1$,

$$\sum_{i=1}^{n} \frac{R(x_i)}{Q'(x_i)} = \text{the coefficient of } x^{n-1} \text{ in } R.$$

In particular the sum is 0 when $\deg R \leq n-2$. For an arbitrary polynomial F, the sum $\sum F(x_i)/Q'(x_i)$ equals the coefficient of x^{n-1} in the remainder of F on division by Q.
Proof. By Theorem 1 (with n nodes), $R = \sum R(x_i) L_i$, and the coefficient of x^{n-1} in L_i is $1/Q'(x_i)$. For the last part, $F = SQ + R$ with $\deg R \leq n-1$, and $F(x_i) = R(x_i)$.

Example 1.
Distinct reals a, b, c, d satisfy $a + b + c + d = 0$. Prove that

$$\frac{a^5}{(a-b)(a-c)(a-d)} + \frac{b^5}{(b-a)(b-c)(b-d)} + \frac{c^5}{(c-a)(c-b)(c-d)} + \frac{d^5}{(d-a)(d-b)(d-c)} = \frac{a^2+b^2+c^2+d^2}{2}.$$

Solution.
The standard way is to put everything over the common denominator, the product of all six differences, and expand a numerator of degree 8. This is possible but long, and the division at the end is not obvious. The denominators are exactly $Q'(a), \ldots, Q'(d)$ for $Q(x) = (x-a)(x-b)(x-c)(x-d)$, so Theorem 2 applies with $F(x) = x^5$. Let e_2, e_3, e_4 be the elementary symmetric functions of a, b, c, d. As $e_1 = 0$, $Q(x) = x^4 + e_2 x^2 - e_3 x + e_4$. Then

$$x^5 = x\,Q(x) - \left(e_2 x^3 - e_3 x^2 + e_4 x\right),$$

so the remainder is $-e_2 x^3 + e_3 x^2 - e_4 x$ and the sum equals $-e_2$. Finally $a^2 + b^2 + c^2 + d^2 = e_1^2 - 2e_2 = -2e_2$, which gives the result.
The same argument with x^m in place of x^5 gives the general identity of Practice 5.

1.2 Finite differences and the degree

Write $\Delta f(x) = f(x+1) - f(x)$, and more generally $\Delta_h f(x) = f(x+h) - f(x)$.

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Theorem 3.
If P has degree $d \geq 1$ and leading coefficient a, then $\Delta_h P$ has degree $d-1$ and leading coefficient $d\,a\,h$. Hence $\Delta_h^d P$ is the constant $d!\,a\,h^d$ and $\Delta_h^{d+1} P = 0$.
Proof. $(x+h)^k - x^k = kh\,x^{k-1} + (\text{lower terms})$, so the term ax^d contributes $dah\,x^{d-1}$ and the other terms contribute only lower powers. Apply this d times.

Theorem 4.
For every function f, every x and every $n \geq 0$,

$$\Delta_h^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+kh), \qquad f(x+nh) = \sum_{k=0}^{n} \binom{n}{k} \Delta_h^k f(x).$$

Proof. Let E be the shift $Ef(x) = f(x+h)$ and I the identity. Then $\Delta_h = E - I$, and E and I commute, so the binomial theorem applies to $(E-I)^n$ and to $E^n = (I + \Delta_h)^n$.

Combining the two theorems gives the identity that solves most problems of this unit.

Corollary 5.
If $\deg P < n$, then $\sum_{k=0}^{n} (-1)^k \binom{n}{k} P(x+kh) = 0$ for all x and h. If $\deg P = n$ with leading coefficient a, then $\sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} P(x+kh) = n!\,a\,h^n$. In particular, if $\deg P \leq n$,

$$P(n+1) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n+1}{k} P(k).$$

Corollary 5 with $h = 1$, $x = 0$ is Theorem 2 for the nodes $0, 1, \ldots, n$, since $\prod_{j \neq k} (k-j) = (-1)^{n-k} k!\,(n-k)!$. Use Theorem 2 when the nodes are arbitrary and Corollary 5 when they are equally spaced.

1.3 Values at integers and two ways to find $P(n+1)$

Problems of the form "P has degree n and $P(k) = v_k$ for $k = 0, 1, \ldots, n$. Find $P(n+1)$" are common, and there are two methods.

1. (The denominator trick.) If $v_k = N(k)/D(k)$ for polynomials N, D, then $Q(x) = D(x)P(x) - N(x)$ is a polynomial with roots $0, 1, \ldots, n$, so $Q(x) = C(x)\,x(x-1)\cdots(x-n)$, where C has degree at most $\deg D - 1$ when $\deg N \leq n + \deg D$. The coefficients of C come from evaluating at the roots of D, where $Q = -N$ (provided these roots are not among $0, \ldots, n$). This needs v_k to be a rational function of k.
2. (The alternating sum.) Corollary 5 gives $P(n+1)$ as a binomial sum of the values, whatever they are. The work is then in evaluating that sum, usually by the binomial theorem or by rewriting $\binom{n+1}{k} v_k$ as a single binomial coefficient.

For values further along, Newton's formula from Theorem 4 gives $P(m) = \sum_{k=0}^{n} \binom{m}{k} \Delta^k P(0)$ for every integer m, which is the difference table of the opening problem written as a formula. Note that "degree n" and "degree at most n" give the same answer here. The values determine P, whatever its degree turns out to be.

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Example 2.
The polynomial P has degree at most n and $P(k) = 2^{-k}$ for $k = 0, 1, \ldots, n$. Find $P(n+1)$.

Solution.
The usual trick for such problems is to multiply by a denominator and obtain a polynomial with known roots, as for values like $\frac{k}{k+1}$. Here the values are not a rational function of k, so the trick has nothing to work on, and solving for the coefficients is hopeless for general n. Evaluate the Lagrange formula at $x = n+1$ instead. For the nodes $0, \ldots, n$,

$$L_k(n+1) = \prod_{j \neq k} \frac{n+1-j}{k-j} = \frac{(n+1)!/(n+1-k)}{(-1)^{n-k} k!\,(n-k)!} = (-1)^{n-k} \binom{n+1}{k},$$

which is the formula of Corollary 5. Hence, by the binomial theorem,

$$P(n+1) = (-1)^n \sum_{k=0}^{n} \binom{n+1}{k} \left(-\tfrac{1}{2}\right)^k = (-1)^n \left[\left(\tfrac{1}{2}\right)^{n+1} - \left(-\tfrac{1}{2}\right)^{n+1}\right] = \frac{1+(-1)^n}{2^{n+1}}.$$

So $P(n+1) = 2^{-n}$ if n is even and $P(n+1) = 0$ if n is odd. As a check, for $n = 1$ the line through $(0, 1)$ and $(1, \frac{1}{2})$ meets the axis at $x = 2$.

1.4 Integer-valued polynomials revisited

L5-AL1 decided which polynomials take integer values at integers. Corollary 5 adds three facts that contest problems use directly.

Theorem 6.
Let P take integer values at all integers and have degree n, and let m be an integer.

1. The highest common factor of all the values $P(k)$, $k \in \mathbb{Z}$, equals the highest common factor of $P(m), P(m+1), \ldots, P(m+n)$.
2. If P has integer coefficients and leading coefficient a, and an integer D divides $P(k)$ for every k, then $D \mid n!\,a$.
3. If p is prime and P has integer coefficients and degree at most $p-2$, then $P(0) + P(1) + \cdots + P(p-1) \equiv 0 \pmod{p}$.

Proof. (1) By Theorem 4, each $\Delta^k P(m)$ is an integer combination of $P(m), \ldots, P(m+n)$. Both sides of $P(m+j) = \sum_{k=0}^{n} \binom{j}{k} \Delta^k P(m)$ are polynomials in j of degree at most n that agree for $j = 0, \ldots, n$, so they agree for every integer j, and $\binom{j}{k}$ is an integer for every integer j (negative ones included). So every value is an integer combination of $P(m), \ldots, P(m+n)$, and conversely. (2) is Corollary 5 with $h = 1$. (3) For $1 \le k \le p-1$, $k!\binom{p-1}{k} = (p-1)(p-2)\cdots(p-k) \equiv (-1)^k k!$, and $k!$ is invertible modulo p, so $\binom{p-1}{k} \equiv (-1)^k$. Corollary 5 with $n = p-1$ gives $0 = \sum (-1)^k \binom{p-1}{k} P(k) \equiv \sum P(k) \pmod{p}$.

Part (2) is often used in the following form. A polynomial with integer coefficients and leading coefficient 1 cannot have all its values divisible by a number that does not divide $n!$. The falling factorial $x(x-1)\cdots(x-n+1)$ shows that $n!$ itself is possible. Also, $n!\,P$ has integer coefficients whenever P is integer-valued of degree n, because $P = \sum_k c_k \binom{x}{k}$ with integers c_k and $n!\binom{x}{k}$ has integer coefficients for $k \le n$. Part (3) contains the classical fact that $1^j + 2^j + \cdots + (p-1)^j \equiv 0 \pmod{p}$ for $1 \le j \le p-2$. For $j = p-1$ the sum is $\equiv -1$ by Fermat, so the bound $p-2$ cannot

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be raised.

Example 3.
Find all pairs of integers (a, b) such that $n^5 + an^3 + bn$ is divisible by 120 for every integer n.

Solution.
The standard way is to work modulo 8, 3 and 5 separately, using $n^5 \equiv n \pmod 5$, $n^3 \equiv n \pmod 3$ and a case split modulo 8. This can be done, but each modulus needs its own argument. By Theorem 6(1) with $m = -2$, the condition holds for every n as soon as the condition holds for $n = -2, -1, 0, 1, 2, 3$. The polynomial P is odd and $P(0) = 0$, so we need exactly

$$120 \mid 1 + a + b, \qquad 120 \mid 32 + 8a + 2b, \qquad 120 \mid 243 + 27a + 3b.$$

Put $b = -1 - a + 120s$. The second condition becomes $120 \mid 30 + 6a$, that is $a \equiv 15 \pmod{20}$, and the third becomes $120 \mid 240 + 24a$, that is $5 \mid a$, which already follows. The answer is $a \equiv 15 \pmod{20}$ and $b \equiv -1 - a \pmod{120}$. For instance $a = -5$, $b = 4$ gives $(n-2)(n-1)n(n+1)(n+2)$.

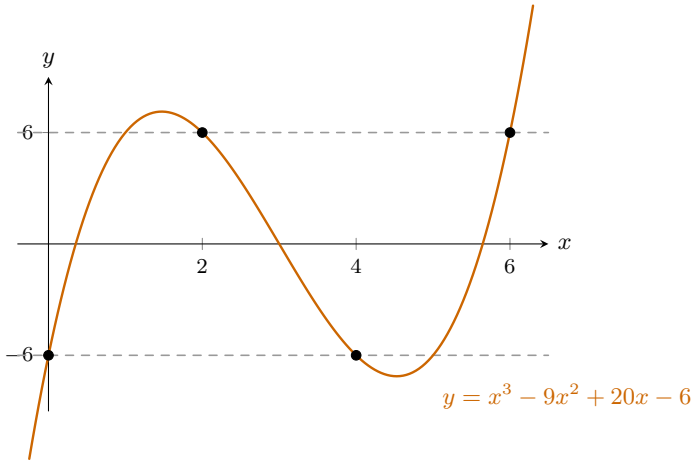
Example 4.
Let P be a polynomial of degree $n \geq 1$ with leading coefficient 1, and let m be a real number. Prove that at least one of $|P(m)|, |P(m+2)|, \ldots, |P(m+2n)|$ is at least $n!$, and that $n!$ cannot be replaced by anything larger.

Solution.
The values at $n+1$ equally spaced points are tied to the leading coefficient by Corollary 5 with $h = 2$:

$$\sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} P(m+2k) = n!\, 2^n.$$

If every $|P(m+2k)|$ were less than $n!$, the triangle inequality would give $n!\, 2^n \leq \sum \binom{n}{k} |P(m+2k)| < n! \sum \binom{n}{k} = n!\, 2^n$, a contradiction. For sharpness, let P be the polynomial of degree at most n with $P(m+2k) = (-1)^{n-k} n!$ for $k = 0, \ldots, n$ (Theorem 1). By Corollary 5 its coefficient of x^n is $\frac{1}{n!\, 2^n} \sum \binom{n}{k} n! = 1$, so the polynomial is monic of degree n, and all the values have absolute value exactly $n!$. The figure shows this polynomial for $n = 3$, $m = 0$.
With spacing 1 the same argument gives the bound $n!/2^n$. The version for arbitrary distinct integer nodes is a well-known olympiad problem. With arbitrary nodes one must use Theorem 2 instead of Corollary 5.

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Common traps.

- Using Corollary 5 without the degree bound. For $\deg P = n+1$ the formula for $P(n+1)$ is wrong by $(n+1)!$ times the leading coefficient. State the bound in your solution.
- Using Corollary 5 at points that are not equally spaced. For arbitrary nodes use Theorem 2.
- Confusing integer values with integer coefficients. For example $\binom{x}{2}$ takes integer values, but $2-0 \mid \binom{2}{2} - \binom{0}{2}$ fails. For integer-valued P of degree n, only $a-b \mid n!\big(P(a)-P(b)\big)$ is true, because $n!P$ has integer coefficients.
- Applying Theorem 6(3) with degree $p-1$. The term x^{p-1} contributes -1, not 0.
- Writing "by Lagrange interpolation" with no formula. Markers want the basis polynomial evaluated at the point you need, or the identity of Corollary 5 stated with its hypothesis.

1.5 How to spot this in a contest

1. "P has degree n and $P(k) = \ldots$ for $k = 0, 1, \ldots, n$. Find $P(n+1)$". If the values are a rational function of k, clear denominators and use the roots. Otherwise use Corollary 5.
2. A sum $\sum F(x_i) / \prod_{j \neq i}(x_i - x_j)$ calls for Theorem 2, after reducing F modulo $\prod(x - x_j)$.
3. An alternating sum $\sum (-1)^k \binom{n}{k} f(k)$ equals $\pm \Delta^n f(0)$, so find the degree of f.
4. "Divides $P(n)$ for every integer n" is reduced by Theorem 6(1) to finitely many values.
5. Bounds on the values of a monic polynomial at many points call for Corollary 5 and the triangle inequality.
6. Polynomials modulo a prime p with degree below $p-1$ call for Theorem 6(3).

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

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Problem 1 ⋆⋆ Let $n \geq 0$ and let P be a polynomial of degree at most n with $|P(k)| \leq 1$ for $k = 0, 1, \ldots, n$. Prove that $|P(n+1)| \leq 2^{n+1} - 1$, and find all P for which equality holds.

Problem 2 ⋆⋆ Find the largest integer that divides $n^5 + 7n^3 + 12n^2 + 4n$ for every integer n.

Problem 3 ⋆⋆⋆ The polynomial P has degree at most n and $P(k) = \frac{1}{(k+1)(k+2)}$ for $k = 0, 1, \ldots, n$. Find $P(n+1)$.

Problem 4 ⋆⋆⋆ Let p be an odd prime and let P be a polynomial with integer coefficients such that $kP(k) \equiv 1 \pmod{p}$ for $k = 1, 2, \ldots, p-1$. Prove that P has degree at least $p-2$, and give an example of degree exactly $p-2$.

Problem 5 ⋆⋆⋆⋆ Let $x_1, \ldots, x_n$ be distinct real numbers, and for $k \geq 0$ let h_k be the sum of all monomials of degree k in $x_1, \ldots, x_n$ (so $h_0 = 1$, $h_1 = \sum x_i$, $h_2 = \sum_{i \leq j} x_i x_j$). Prove that for every $k \geq 0$,

$$\sum_{i=1}^{n} \frac{x_i^{n-1+k}}{\prod_{j \neq i}(x_i - x_j)} = h_k.$$

Problem 6 ⋆⋆⋆⋆ Let p be an odd prime and let P be a polynomial with integer coefficients of degree d, where $d \geq 2$, d divides $p-1$, and the leading coefficient of P is not divisible by p. Prove that the numbers $P(0), P(1), \ldots, P(p-1)$ are not pairwise distinct modulo p.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO Round 1 2015/16 Problem 3** ⋆⋆
   A quadratic sequence has constant second differences. Compare with the differences of consecutive Fibonacci numbers.
2. **USAMO 1975 Problem 3** ⋆⋆
   Clear the denominator to get a polynomial of degree n + 1 whose roots you know.
3. **IMO Shortlist 1981 Problem 6** ⋆⋆
   Use the alternating binomial sum over the n + 2 points 0, 1, ..., n + 1.
4. **USAMO 1984 Problem 5** ⋆⋆⋆
   Apply the alternating binomial identity to the 3n + 2 points 0, 1, ..., 3n + 1 and split the sum by residues mod 3.
5. **IMO Shortlist 1983 Problem 18** ⋆⋆⋆
   Differences of Fibonacci numbers are Fibonacci numbers. Use Newton's forward formula starting from k = 992.
6. **IMO Shortlist 1997 Problem 12** ⋆⋆⋆
   Suppose the degree is at most p - 2 and add the values over a complete residue system modulo p.
7. **USAMO 2002 Problem 3** ⋆⋆⋆
   Prescribe the values of one of the two polynomials at n points, with alternating signs, and interpolate.
8. **IMO Shortlist 1996 A5** ⋆⋆⋆⋆
   Express the coefficients through the values at 1, -1, 1/2 and -1/2.

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9. **IMO Shortlist 2019 A5** ⋆⋆⋆⋆

Use the leading-coefficient form of Lagrange interpolation for a well-chosen polynomial, possibly with extra nodes.

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4 Hints

1. Write $P(n+1)$ in terms of $P(0), \ldots, P(n)$ by Corollary 5.
2. By Theorem 6(1), six consecutive values decide the answer.
3. Use Corollary 5, and write $\binom{n+1}{k}\frac{1}{(k+1)(k+2)}$ as a single binomial coefficient divided by $(n+2)(n+3)$.
4. If the degree were at most $p-3$, Theorem 6(3) would apply to $xP(x)$.
5. Show that both sides satisfy the same linear recurrence of order n, and compare the first n terms using Theorem 2.
6. If they were distinct, they would be $0, 1, \ldots, p-1$ in some order, so their jth powers would have the same sum modulo p as $0^j, 1^j, \ldots, (p-1)^j$. Choose j so that P^j has degree exactly $p-1$.

5 Solutions

Problem 1

Let $n \geq 0$ and let P be a polynomial of degree at most n with $|P(k)| \leq 1$ for $k = 0, 1, \ldots, n$. Prove that $|P(n+1)| \leq 2^{n+1} - 1$, and find all P for which equality holds.

By Corollary 5, $P(n+1) = \sum_{k=0}^{n} (-1)^{n-k}\binom{n+1}{k}P(k)$. By the triangle inequality,

$$|P(n+1)| \leq \sum_{k=0}^{n} \binom{n+1}{k}|P(k)| \leq \sum_{k=0}^{n} \binom{n+1}{k} = 2^{n+1} - 1.$$

Equality in the second step needs $|P(k)| = 1$ for every k, and equality in the first needs the numbers $(-1)^{n-k}P(k)$ to have the same sign, since every $\binom{n+1}{k}$ is positive. So equality holds if and only if $P(k) = \varepsilon(-1)^{n-k}$ for $k = 0, \ldots, n$, with a fixed $\varepsilon \in \{1, -1\}$. By Theorem 1 each choice of ε gives exactly one polynomial of degree at most n, and for this polynomial $P(n+1) = \varepsilon(2^{n+1} - 1)$. So there are exactly two extremal polynomials, P_0 and $-P_0$, where P_0 is the polynomial of degree at most n with $P_0(k) = (-1)^{n-k}$ for $k = 0, \ldots, n$.

Problem 2

Find the largest integer that divides $n^5 + 7n^3 + 12n^2 + 4n$ for every integer n.

Let $P(n) = n^5 + 7n^3 + 12n^2 + 4n$. This polynomial has integer coefficients and degree 5, so by Theorem 6(1) the highest common factor of all its values equals that of $P(0), \ldots, P(5)$. These are

$$0, \quad 24, \quad 144, \quad 552 = 24 \cdot 23, \quad 1680 = 24 \cdot 70, \quad 4320 = 24 \cdot 180,$$

and their highest common factor is 24, since $P(1) = 24$ and the others are multiples of 24. So the answer is 24. (Theorem 6(2) says in advance that the answer divides $5! = 120$. Factorising gives only $P(n) = n(n+1)(n^3 - n^2 + 8n + 4)$, which explains a factor 2 but not 24.)

Answer: 24

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Problem 3

The polynomial P has degree at most n and $P(k) = \frac{1}{(k+1)(k+2)}$ for $k = 0, 1, \ldots, n$. Find $P(n+1)$.

By Corollary 5, $P(n+1) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n+1}{k} \frac{1}{(k+1)(k+2)}$. Now

$$\binom{n+1}{k} \frac{1}{(k+1)(k+2)} = \frac{(n+1)!}{(k+2)!\,(n+1-k)!} = \frac{1}{(n+2)(n+3)} \binom{n+3}{k+2}.$$

Put $j = k+2$, so $(-1)^{n-k} = (-1)^n (-1)^j$, and

$$P(n+1) = \frac{(-1)^n}{(n+2)(n+3)} \sum_{j=2}^{n+2} (-1)^j \binom{n+3}{j}.$$

The full alternating sum $\sum_{j=0}^{n+3} (-1)^j \binom{n+3}{j}$ is 0, and the missing terms $j = 0, 1, n+3$ contribute $1 - (n+3) + (-1)^{n+3}$, so the sum from 2 to $n+2$ is $n + 2 + (-1)^n$. Hence

$$P(n+1) = \frac{(-1)^n \big(n + 2 + (-1)^n\big)}{(n+2)(n+3)} = \frac{(-1)^n (n+2) + 1}{(n+2)(n+3)}.$$

Check. $n = 0$ gives $\frac{1}{2}$ (the constant $P = \frac{1}{2}$), and $n = 1$ gives $-\frac{1}{6}$, the value at 2 of the line through $(0, \frac{1}{2})$ and $(1, \frac{1}{6})$. The denominator trick also works here, since $(x+1)(x+2)P(x) - 1$ has degree at most $n+2$ and roots $0, \ldots, n$, so the polynomial equals $(\alpha x + \beta) x (x-1) \cdots (x-n)$, and $x = -1, -2$ determine α, β. This gives the same answer with more algebra.

Answer: $\frac{(-1)^n (n+2)+1}{(n+2)(n+3)}$

Problem 4

Let p be an odd prime and let P be a polynomial with integer coefficients such that $kP(k) \equiv 1 \pmod{p}$ for $k = 1, 2, \ldots, p-1$. Prove that P has degree at least $p-2$, and give an example of degree exactly $p-2$.

P is not the zero polynomial, since $1 \cdot P(1) \equiv 1$. Suppose $\deg P \le p-3$. Then $Q(x) = xP(x)$ has integer coefficients and degree at most $p-2$, so by Theorem 6(3),

$$Q(0) + Q(1) + \cdots + Q(p-1) \equiv 0 \pmod{p}.$$

But $Q(0) = 0$ and $Q(k) = kP(k) \equiv 1$ for $1 \le k \le p-1$, so the sum is $\equiv p-1 \equiv -1 \pmod{p}$, a contradiction. Hence $\deg P \ge p-2$.

For the example take $P(x) = x^{p-2}$. Then $kP(k) = k^{p-1} \equiv 1 \pmod{p}$ for $1 \le k \le p-1$ by Fermat's little theorem. (So $P(k)$ is the inverse of k modulo p, and no polynomial of smaller degree computes inverses modulo p.)

Problem 5

Let $x_1, \ldots, x_n$ be distinct real numbers, and for $k \ge 0$ let h_k be the sum of all monomials of degree k in $x_1, \ldots, x_n$ (so $h_0 = 1$, $h_1 = \sum x_i$, $h_2 = \sum_{i \le j} x_i x_j$). Prove that for every $k \ge 0$,

$$\sum_{i=1}^{n} \frac{x_i^{n-1+k}}{\prod_{j \ne i} (x_i - x_j)} = h_k.$$

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Let $Q(x) = \prod_j (x - x_j) = x^n - e_1 x^{n-1} + e_2 x^{n-2} - \cdots + (-1)^n e_n$, where e_j are the elementary symmetric functions, and let $T_m = \sum_i x_i^m / Q'(x_i)$ for $m \geq 0$. Recall $Q'(x_i) = \prod_{j \neq i}(x_i - x_j)$. We prove $T_m = h_{m-n+1}$ for all $m \geq 0$, where $h_k = 0$ for $k < 0$.

The recurrence for T. Since $Q(x_i) = 0$, for $m \geq n$ we have $x_i^m = x_i^{m-n}\big(e_1 x_i^{n-1} - e_2 x_i^{n-2} + \cdots + (-1)^{n+1} e_n\big)$. Dividing by $Q'(x_i)$ and adding over i,

$$T_m = \sum_{j=1}^{n} (-1)^{j+1} e_j T_{m-j} \qquad (m \geq n).$$

The recurrence for h. As formal power series in t, $\prod_j \frac{1}{1 - x_j t} = \prod_j \left(1 + x_j t + x_j^2 t^2 + \cdots\right) = \sum_{k \geq 0} h_k t^k$, because expanding the product produces every monomial of degree k exactly once as a coefficient of t^k. Multiplying by $\prod_j (1 - x_j t) = \sum_{j=0}^{n} (-1)^j e_j t^j$ gives 1, so comparing coefficients of t^k for $k \geq 1$,

$$\sum_{j=0}^{n} (-1)^j e_j h_{k-j} = 0, \qquad \text{that is} \qquad h_k = \sum_{j=1}^{n} (-1)^{j+1} e_j h_{k-j} \qquad (k \geq 1).$$

Put $u_m = h_{m-n+1}$. For $m \geq n$ we have $k = m - n + 1 \geq 1$, so $u_m = \sum_{j=1}^{n} (-1)^{j+1} e_j u_{m-j}$. The sequences (T_m) and (u_m) satisfy the same recurrence for $m \geq n$.

The first n terms. By Theorem 2 applied to $R(x) = x^m$ with $0 \leq m \leq n-1$, $T_m = 0$ for $m \leq n-2$ and $T_{n-1} = 1$. Also $u_m = h_{m-n+1} = 0$ for $m \leq n-2$ and $u_{n-1} = h_0 = 1$.

Two sequences that agree in their first n terms and satisfy the same recurrence of order n agree everywhere, by induction on m. Hence $T_{n-1+k} = u_{n-1+k} = h_k$ for every $k \geq 0$. Example 1 is the case $n = 4$, $k = 2$, where $h_2 = e_1^2 - e_2$ and $e_1 = 0$.

Problem 6

Let p be an odd prime and let P be a polynomial with integer coefficients of degree d, where $d \geq 2$, d divides $p - 1$, and the leading coefficient of P is not divisible by p. Prove that the numbers $P(0), P(1), \ldots, P(p-1)$ are not pairwise distinct modulo p.

Suppose the values are pairwise distinct modulo p. There are p of them, so modulo p they are $0, 1, \ldots, p-1$ in some order, and for every $j \geq 1$,

$$\sum_{k=0}^{p-1} P(k)^j \equiv \sum_{k=0}^{p-1} k^j \pmod{p}.$$

Let $j = \frac{p-1}{d}$, an integer with $1 \leq j \leq \frac{p-1}{2} \leq p - 2$, as $d \geq 2$ and $p \geq 3$. By Theorem 6(3) applied to x^j, the right side is $\equiv 0$.

Let a be the leading coefficient of P. Then $P(x)^j = a^j x^{p-1} + R(x)$ with R an integer polynomial of degree at most $p - 2$. By Theorem 6(3), $\sum_k R(k) \equiv 0$, and by Fermat's little theorem $\sum_{k=0}^{p-1} k^{p-1} \equiv p - 1 \equiv -1$, since $k^{p-1} \equiv 1$ for $k \neq 0$. Hence the left side is $\equiv -a^j \not\equiv 0 \pmod{p}$, because $p \nmid a$. This contradiction proves the claim.

Both hypotheses on d are needed. For $d = 1$, $P(x) = x$ gives distinct values, and for $p = 5$, $P(x) = x^3$ (with $3 \nmid 4$) gives $0, 1, 3, 2, 4$, which are distinct.

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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